

Statistics Unit 3: Simple Contrasts Lecture Notes

Three types of studies about means: First Type Single sample

	x	Linear equations	x_{is} and e_{is} are variables Variables have variance within a sample (people can have different values) $\bar{x}_s$ is a parameter estimate
	7		
	12	Variance in a <i>sample</i> $x_{is} = \bar{x}_s + e_{is}$	
	9		
	10	Sampling error variance in a <i>population</i> $\bar{x}_{sp} = \mu_p + e_{sp}$	
	12		

$$\bar{x}_s = 10$$

For case 1 if H_0 true.

$$7 = 10 + (-3)$$

$$10 = 0 + 10$$

Two types of sd

$$sd = sd \text{ of } e_{is}$$

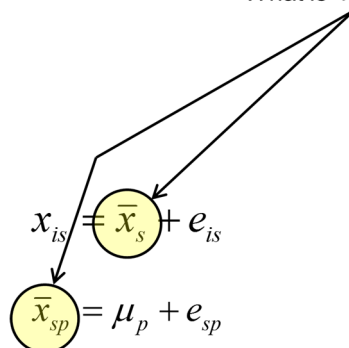
$$sd_{\bar{x}} = sd \text{ of } e_{sp}$$

Two key questions

Descriptive

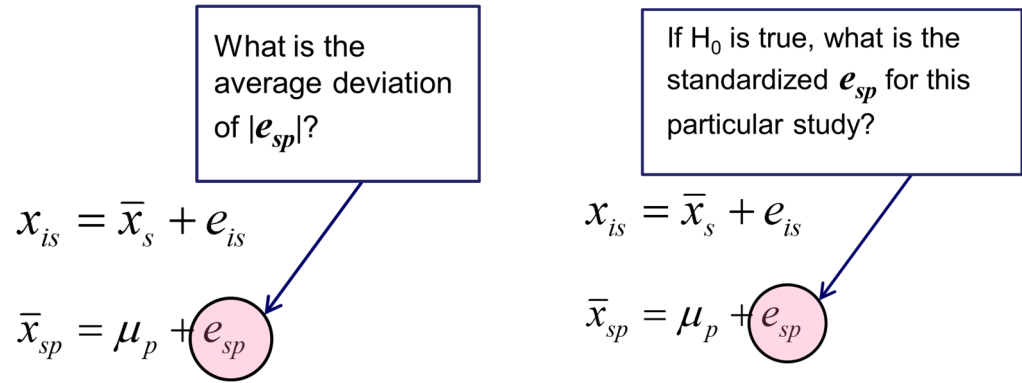
What is the parameter estimate?

What is $\bar{x}$?



Inferential

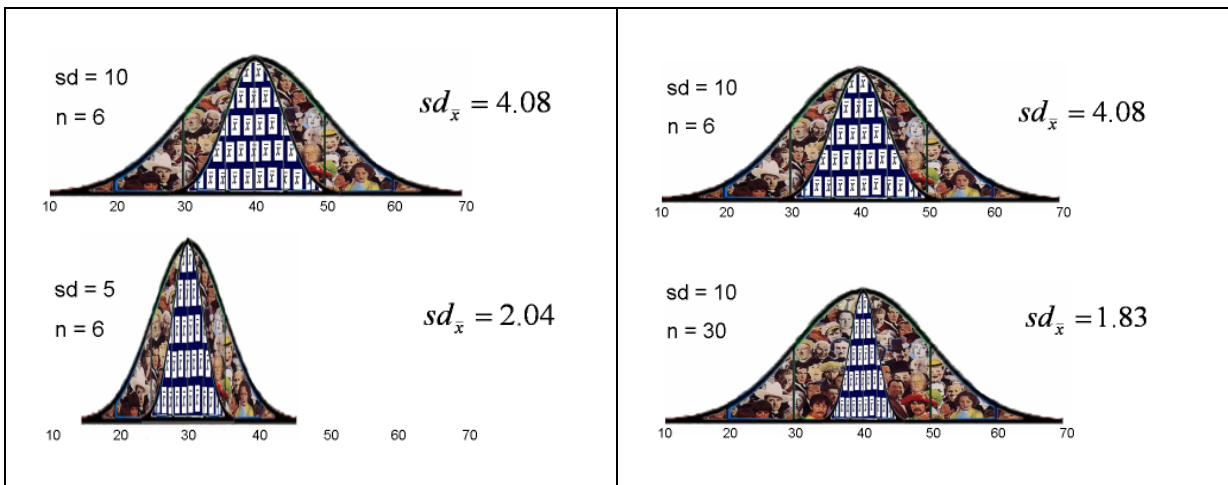
How well does the parameter estimate describe the population?



$z = \frac{x - \bar{x}}{sd}$ z-score: a standardized e_{is} for a single person $x_{is} = \bar{x}_s + e_{is}$ How many standard deviation units did this person deviate from the sample mean? What is the probability of randomly selecting a person with a score this high?	$t = \frac{\text{statistic} - \text{parameter}}{sd_{\text{statistic}}}$ t-value: a standardized e_{sp} for a single study $\bar{x}_{sp} = \mu_p + e_{sp}$ How many standard error units did this study deviate from the null population parameter? What is the probability of randomly selecting a sample with a statistic this high?
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Calculating a Single Sample t -test

Statistic $\bar{x}$ Parameter $\mu = 0$ $t = \frac{\bar{x} - 0}{sd_{\bar{x}}} = \frac{\bar{x}}{sd_{\bar{x}}}$	if : $\mu = 0$ $t = \frac{\bar{x}}{sd_{\bar{x}}}$ $sd_{\bar{x}} = \frac{sd}{\sqrt{n}}$ $df = n - 1$
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	x
	7
	12
	9
	10
	12

descriptive

$$\bar{x}_s = 10 \quad sd = 2.12$$

Linear equations

$$x_{is} = \bar{x}_s + e_{is}$$

$$\bar{x}_{sp} = \mu_p + e_{sp}$$

$$sd_{\bar{x}} = \frac{2.12}{\sqrt{5}} = .95$$

$$\text{inferential } t = \frac{10}{.95} = 10.53$$

Three types of studies about means: Second Type

Paired Samples, Matched samples, Repeated measure, Difference scores

diff			$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$	$\bar{x}_{\text{diff}} = 2 \quad sd_{\text{diff}} = 1$
x_1	x_2	$x - x$	$\bar{x}_1 = 6.2 \quad sd_1 = 2.6$ $\bar{x}_2 = 4.2 \quad sd_2 = 2.6$	Linear equations $x_{\text{diff } is} = \bar{x}_{\text{diff } s} + e_{\text{diff } is}$ $\bar{x}_{\text{diff } sp} = \mu_{\text{diff } p} + e_{\text{diff } sp}$ For case 1 if H_0 true. $1 = 2 + (-1)$ $2 = 0 + 2$
9	8	1		
8	5	3		
7	4	3		
4	3	1		
3	1	2		

Identify:

1. Equation for variance in a sample
2. Equation for sampling error variance in a population
3. Variables
4. Parameter estimate

Two key questions

Descriptive

What is the parameter estimate?

What is $\bar{x}_{\text{diff}}$?

Inferential

How well does the parameter estimate describe the population?

What is the average deviation of

 $|e_{\text{diff } sp}|$?If H_0 is true, what is thestandardized $e_{\text{diff } sp}$ for this study?

Statistic	Parameter	
$\bar{x}_{\text{difference}}$	$\mu_{\text{difference}} = 0$	if : $\mu_{\text{difference}} = 0 \quad t = \frac{\bar{x}_{\text{difference}}}{sd_{\bar{x} \text{ difference}}}$
$t = \frac{\bar{x}_{\text{difference}} - 0}{sd_{\bar{x} \text{ difference}}} = \frac{\bar{x}_{\text{difference}}}{sd_{\bar{x} \text{ difference}}}$		$sd_{\bar{x} \text{ difference}} = \frac{sd_{\text{difference scores}}}{\sqrt{n}}$
		$df = n - 1$

diff		
x_1	x_2	$x - x$
9	8	1
8	5	3
7	4	3
4	3	1
3	1	2

descriptive

$$\bar{x}_{\text{diff}} = 2$$

$$sd_{\text{diff}} = 1$$

Linear equations

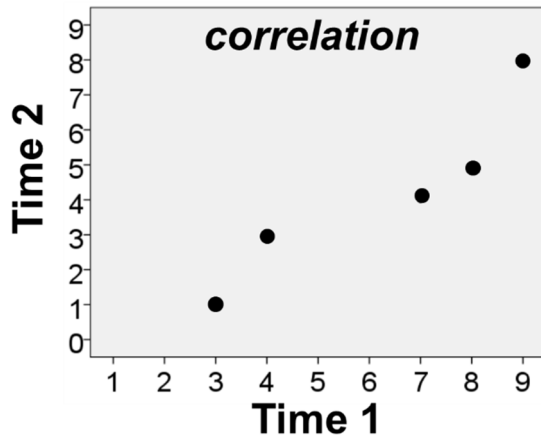
$$x_{\text{diff } is} = \bar{x}_{\text{diff } s} + e_{\text{diff } is}$$

$$\bar{x}_{\text{diff } sp} = \mu_{\text{diff } p} + e_{\text{diff } sp}$$

$$sd_{\bar{x} \text{ difference}} = \frac{1}{\sqrt{5}} = .447$$

$$\text{inferential} \quad t = \frac{2}{.447} = 4.47$$

Paired sample correlation



Three types of studies about means: Third Type
Two Samples, Independent samples, Two groups

<table border="1"> <thead> <tr> <th>y</th><th>x</th></tr> </thead> <tbody> <tr><td>3</td><td>1</td></tr> <tr><td>4</td><td>1</td></tr> <tr><td>2</td><td>1</td></tr> <tr><td>3</td><td>1</td></tr> <tr><td>8</td><td>0</td></tr> <tr><td>8</td><td>0</td></tr> <tr><td>7</td><td>0</td></tr> <tr><td>9</td><td>0</td></tr> </tbody> </table> $y =$ level of happiness $x =$ group membership (1 or 0) <i>Note:</i> in this unit group 1 will be listed first group 0 will be listed second	y	x	3	1	4	1	2	1	3	1	8	0	8	0	7	0	9	0	$\bar{y}_1 = 3$ $\bar{y}_0 = 8$ $\bar{y}_1 - \bar{y}_0 = -5$ $y_i = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x_i + e$ Group 0 mean $\hat{y}_i = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x_i$ Difference between groups Expected increase for group 1 If group = 1 $\hat{y}_i = 8 + (-5)1 = \bar{y}_1 = 3$ If group = 0 $\hat{y}_i = 8 + (-5)0 = \bar{y}_0 = 8$
y	x																		
3	1																		
4	1																		
2	1																		
3	1																		
8	0																		
8	0																		
7	0																		
9	0																		

Linear equations

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

Identify:

1. Equation for variance in a sample
2. Equation for sampling error variance in a population
3. Variables
4. Parameter estimates

For case 2 if H_0 true	$4 = 8 + (-5)(1) + 1$ $8 = 0 + 8$ $-5 = 0 + (-5)$	For case 7 if H_0 true	$7 = 8 + (-5)(0) + (-1)$ $8 = 0 + 8$ $-5 = 0 + (-5)$
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Two key questions

Descriptive

What is the parameter estimate?

$$y_{is} = \bar{y}_{0s} + (\bar{y}_{1s} - \bar{y}_{0s})x_{is} + e_{is}$$

$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

$\bar{y}_1 - \bar{y}_0$?

Inferential

How well does the parameter estimate describe the population?

What is the average deviation of $|e_{1-0sp}|$?

If H_0 is true, what is the standardized e_{1-0sp} for this study?

~~$$\bar{y}_{0sp} = \mu_{0p} + e_{0sp}$$~~

$$(\bar{y}_{1sp} - \bar{y}_{0sp}) = (\mu_{1p} - \mu_{0p}) + e_{1-0sp}$$

Statistic	Parameter	
$\bar{y}_1 - \bar{y}_0$	$\mu_1 - \mu_0 = 0$	if : $\mu_1 - \mu_0 = 0 \quad t = \frac{\bar{y}_1 - \bar{y}_0}{sd_{\bar{y}-\bar{y}}}$
$t = \frac{(\bar{y}_1 - \bar{y}_0) - 0}{sd_{\bar{y}-\bar{y}}} = \frac{\bar{y}_1 - \bar{y}_0}{sd_{\bar{y}-\bar{y}}}$		$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{sd_{pooled}^2}{n_1} + \frac{sd_{pooled}^2}{n_2}}$ $df = (n_1 - 1) + (n_2 - 1)$ $df = n_{total} - 2$

$$ss = \sum (y - \bar{y})^2$$

$$sd^2 = \frac{ss}{df}$$

$$sd_{pooled}^2 = \frac{ss_1 + ss_2}{df_1 + df_2}$$

$$sd = \sqrt{\frac{ss}{df}}$$

$$sd_{pooled} = \sqrt{\frac{ss_1 + ss_2}{df_1 + df_2}}$$

y	x
3	1
4	1
2	1
3	1
8	0
8	0
7	0
9	0

$$\bar{y}_1 = 3$$

$$ss_1 = 2$$

$$sd_1^2 \approx .67$$

$$sd_1 \approx .82$$

$$\bar{y}_0 = 8$$

$$ss_0 = 2$$

$$sd_0^2 \approx .67$$

$$sd_0 \approx .82$$

$$sd_{pooled}^2 = \frac{ss_1 + ss_2}{df_1 + df_2} = \frac{2 + 2}{3 + 3} \approx .667$$

$$\bar{y}_1 - \bar{y}_0 = -5$$

$$t = \frac{-5}{.58} \approx -8.62$$

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{sd_{pooled}^2}{n_1} + \frac{sd_{pooled}^2}{n_2}} = \sqrt{\frac{.667}{4} + \frac{.667}{4}} \approx .58$$

Descriptive statistics: Standardized Effect Sizes

Cohen's d

Point biserial correlation

$$d = \frac{\bar{y}_1 - \bar{y}_0}{sd_{pooled}}$$

$$r_{pb} = \sqrt{\frac{t^2}{t^2 + df}}$$

if: $n_1 = n_2$

$$sd_{pooled} = \sqrt{\frac{sd_1^2 + sd_0^2}{2}}$$

Assumptions for the t-value

Heterogeneity of variance

Assumption: population variances are equal

Levene's test for equality of variances

Independence of observations

Advanced Topic that is NOT part of this course

Information to consider when calculating paired sample effect sizes:

Connections between the sd of difference scores, the sd of individual scores, and the paired sample r .

Above, I provided instructions to calculate a paired sample d -statistic using the pooled standard deviations for the **individual scores**. However, another option for calculating a d -statistic is to use the standard deviation of the **difference score**. This other option is more popular, but when this option is used, it is important to know that the size of the d -statistic will be influenced by the size of the paired sample correlation (note that I am talking about the paired sample correlation, the extent of stability in the rank order of cases across two assessments, and NOT the point-biserial correlation). This is because there is a connection between the paired sample correlation and the standard deviation of a difference score. Specifically, the standard deviation of a difference score (sd_{ds}) is inversely associated with the paired sample correlation (r_{ps}). The greater the stability in the rank order of cases across two assessments, the smaller the standard deviation of the difference score.

sd_{ds} can be computed from r_{ps} , sd_{x1} , and sd_{x2} :
$$sd_{ds} = \sqrt{sd_{x1}^2 + sd_{x2}^2 - (2)(r_{ps})(sd_{x1})(sd_{x2})}$$

(Note: In using this formula, keep in mind the fact that sd^2 is a variance and sd is a standard deviation.)

The table below indicates the size of a standard deviation for a difference score (sd_{ds}) based on the paired sample correlation (r_{ps}) assuming $sd_{x1} = 1$ and $sd_{x2} = 1$.

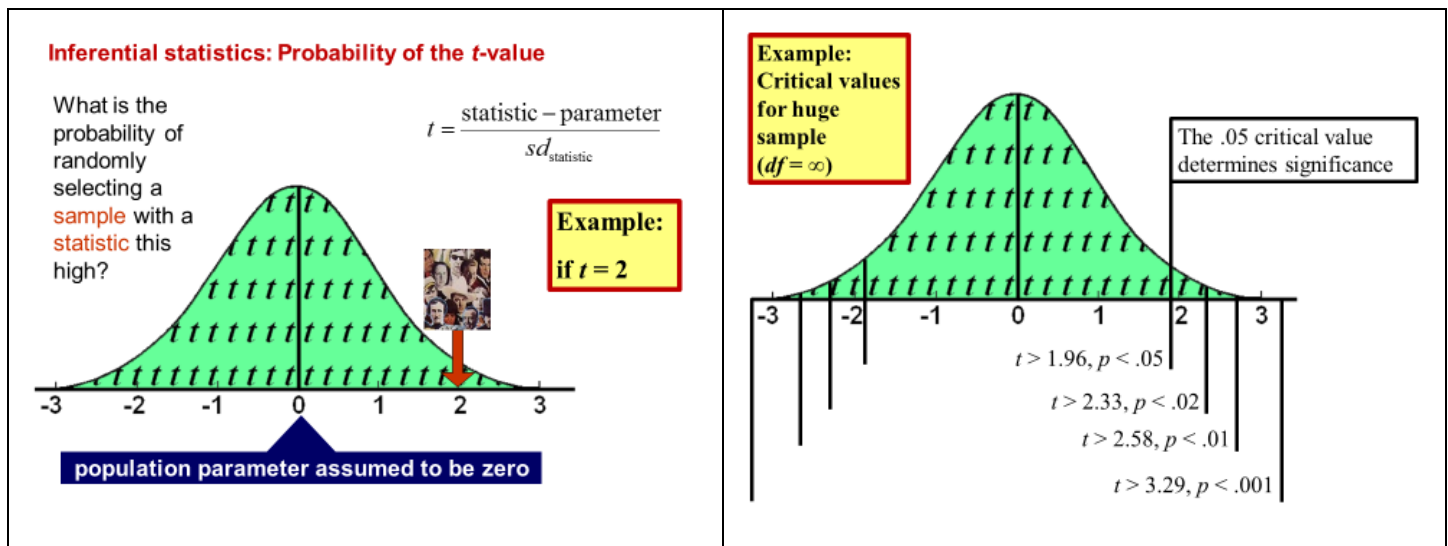
r_{ps}	sd_{ds}
0	1.414
.1	1.342
.2	1.265
.3	1.183
.4	1.095
.5	1.000
.6	0.894
.7	0.775
.8	0.632
.9	0.447
1.0	0

What does all this mean?

In comparing d -statistics based on pooled individual score sds versus difference score sds :

- If $r_{ps} = .5$, then both approaches will use the same sd value, which means they will both produce the same d -statistic.
- As r_{ps} increases, the d -statistic based on sd_{ds} will also increase (because as the as r_{ps} increases the, the sd_{ds} shrinks, and when the sd shrinks, the d will increase).

Calculating a p -value

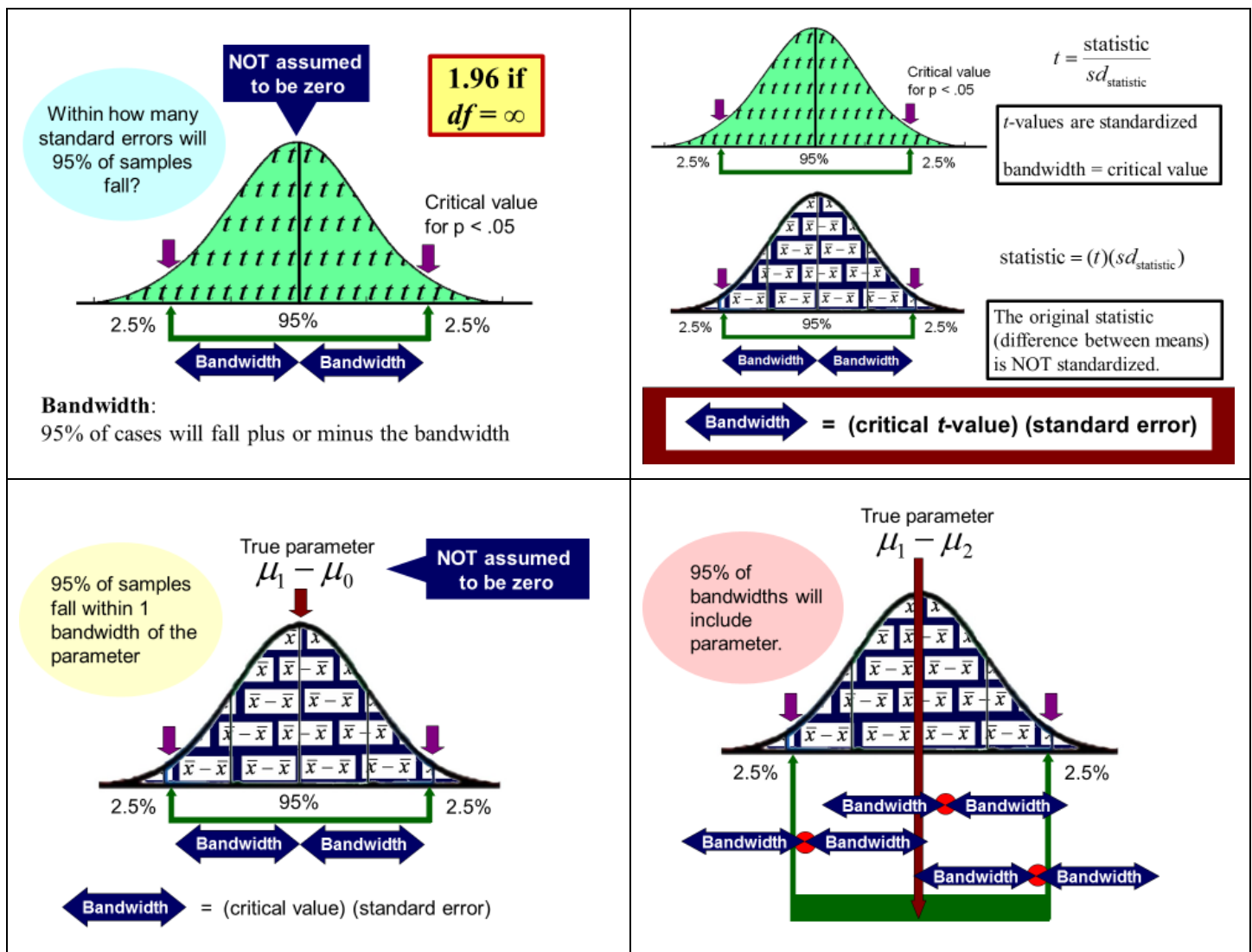


Need the t -value and the df

Option 1: Use computer (or app) and calculate precise value

Option 2: Use printed table and look up critical values

Confidence intervals



$$95\% \text{ CI} = \text{Statistic} \pm \text{bandwidth}$$

$$\text{Bandwidth} = (\text{critical value}) (\text{standard error})$$

Five Pieces of Information

Descriptive statistic (size of effect)

1. Name
2. Value

Inferential statistic

3. Name
4. Value
5. Probability (enter exactly as it appears)

Lecture Add On Notes

Premarital Education and Commitment

Stanley, Amato, Johnson, and Markman (2006). Premarital education, marital quality, and marital stability: Findings from a large, random household survey. *Journal of Family Psychology*, 20, 117-126.

Did you have premarital education?

Commitment : 3 items rated on 5-point scale (Alpha = .69)

1. My relationship is more important to me than almost anything else in my life.
2. I may not want to be with my spouse a few years from now (reverse scored).
3. I like to think of my spouse and me more in terms of us and we than me and him/her.

Group Statistics

group	N	Mean	Std. Deviation	Std. Error Mean
commitment no ed	1365	11.12	2.50	.07
ed	612	11.64	2.39	.10

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
commitment	Equal variances assumed	1.04	.31	-4.31	1975	.000	-.52	.12	-.75	-.28
	Equal variances not assumed			-4.39	1225.75	.000	-.52	.12	-.75	-.29

Sample Homework Questions

Computations

1. Standard error of mean
2. Mean difference
3. Pooled variance
4. Standard error of difference
5. Degrees of freedom
6. *t*-value
7. *p* value
8. *d* statistic (need the pooled standard deviation)
9. point-biserial correlation
10. Compute the 95% confidence interval.

Two group questions:

1. In group X, what is the distance that people's scores typically deviated from the group mean?
2. If this study were repeated over and over, how much would you typically expect the mean for group X to deviate from the true population mean for this group?
3. If this study were repeated over and over, how much would you typically expect the size of difference between the two groups to deviate from the true difference in the population?
4. Based on the linear equations discussed in class, what is the standardized $e_{1-0\ sp}$ for this study?
5. Based on the linear equations discussed in class, what is the average deviation for $|e_{1-0\ sp}|$?

Is there a difference between groups? *To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.*

1. the name of the descriptive statistic
2. the value of the descriptive statistic
3. the name of the inferential statistic
4. the value of the inferential statistic
5. the p-value (**report exactly as printed with NO rounding**)
6. This effect is best described as:

Written summary

There was a significant difference between the couples that had premarital education and the couples that did not ($d = .21$, $t(1,975) = 4.31$, $p < .001$), with the premarital education couples reporting more commitment ($M = 11.64$, $SD = 2.39$) than the couples receiving no premarital education ($M = 11.12$, $SD = 2.50$). This represents a small effect.

Round numbers to two decimals

If $p < .01$, round p -value to 3 decimals

If $p < .001$, print $p < .001$

NEVER print $p = .000$

The homework for this unit provides details on how written summaries are scored.

Demand-Withdraw Communication

Based on data drawn from: Sanford, K. (2007). The Couples Emotion Rating Form: Psychometric properties and theoretical associations. *Psychological Assessment*, 19, 411-421.

Communication Patterns Questionnaire

Sample item: Woman nags and demands while man withdraws, becomes silent, or refuses to discuss the matter further.

Wife Withdrawal	Husband Withdrawal
6 items total	6 items total
3 completed by wife	3 completed by wife
3 completed by husband	3 completed by husband
Alpha = .73	Alpha = .78

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	wife_withdraw	3.57	394	1.62	.08
	husb_withdraw	4.53	394	1.79	.09

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	wife_withdraw & husb_withdraw	394	.18	.000

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	wife_withdraw - husb_withdraw	-.96	2.19	.11	-1.18	-.75	-8.73	393	.000

Paired sample questions:

1. What is the average difference score?
2. At time X (or for partner X), what is the distance that people's scores typically deviated from the group mean?
3. What is the typical distance that people's difference scores deviated from the group mean difference score?
4. If this study were repeated over and over, how much would you typically expect the mean for time X (or for partner X) to deviate from the true population mean?
5. If this study were repeated over and over, how much would you typically expect the size of group mean difference score to deviate from the true difference in the population?
6. Calculate the t -value
7. Calculate the d statistic
8. What is the numerical value indicates the degree of association between scores at time 1 and scores at time 2 (or association between partner 1 and partner 2)?
9. The degree of association between scores at time 1 and scores at time 2 is best described as:

Written summary

There was a significant difference between wives and husbands ($d = .56$, $t(393) = 8.73$, $p < .001$), with the wives receiving lower withdrawal scores ($M = 3.57$, $SD = 1.62$) than husbands ($M = 4.53$, $SD = 1.79$). This represents a medium effect.

The homework for this unit provides details on how written summaries are scored.

Additional Writing Practice

This page includes additional examples you may use for practice in writing written summaries.

Married people were asked to think of a recent conflict in their marriage. Each participant rated the extent to which he or she perceived soft emotion (feeling sad, hurt, disappointed, concerned) in the partner at the time of the conflict. Participants also indicated whether or not the conflict was resolved.

Group Statistics

		N	Mean	Std. Deviation	Std. Error Mean
p_soft	resolved no	119	1.67	1.10	.10
	yes	564	2.04	.94	.04

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
p_soft	Equal variances assumed	5.593	.018	-3.699	681	.000	-.361	.098	-.552	-.169
	Equal variances not assumed			-3.322	155.716	.001	-.361	.109	-.575	-.146

The following results are from married couples in which both the wife and the husband rated the extent of perceived soft emotion in the partner. Is there a difference between wives and husbands?

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	wsp_soft	1.931	84	1.388	.151
	hsp_soft	2.857	84	1.512	.165

Paired Samples Correlations

		N	Correlation	Sig.
Pair 1	wsp_soft & hsp_soft	84	.036	.742

Paired Samples Test

		Paired Differences					t	df	Sig. (2-tailed)
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper			
Pair 1	wsp_soft - hsp_soft	-.927	2.014	.220	-1.364	-.489	-4.216	83	.000